

Barium Titanate DP-IQM Enabling Net 1 Tbps/ λ ZR and Coherent-Lite Data Center Networks

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Abstract: We present a barium titanate (BTO) dual polarization in-phase quadrature modulator (DP-IQM) enabling a record net 1 Tbps/ λ over 80 km in a ZR modeled system and over 2 km in a coherent-lite system.

1. Introduction

Compared to IM/DD, coherent solutions offer superior receiver sensitivity and spectral efficiency, however, their higher power consumption, complexity, and cost have traditionally limited deployment to long-reach applications such as OIF ZR links [1]. As AI-driven data center traffic and lane rates increase, extending cost-effective IM/DD systems beyond 200 Gbps/lane becomes increasingly challenging [2]. Coherent-lite architectures are emerging as a practical alternative for short-reach links, particularly as the power per bit of coherent transmission approaches that of IM/DD with scaling the of data rates [3]. One architecture is the bi-directional self-homodyne coherent system, which leverages existing duplex fibers in data centers [4]. Using the same laser for both the carrier and local oscillator (LO) simplifies receiver DSP by removing frequency offset (FO) estimation and carrier recovery, while chromatic dispersion (CD) can be mitigated by using shorter fiber spans. Eliminating carrier recovery, FO, and CD compensation DSP enables an 800 G LR transceiver to reduce power consumption by 14 % [5]. Furthermore, because self-homodyne systems can operate with higher linewidth lasers, DFB lasers can be deployed with minimal performance degradation [4], for a cost-effective solution.

To date, no full dual-polarization (DP) transmission system based on a BTO DP-IQM has been demonstrated. Previous BTO-enabled transmission systems have employed MZMs [6], ring modulators [7] and plasmonic modulators [8]. BTO modulators are a new technology for high-speed optical communications offering a large Pockels coefficient, low optical loss and integration in silicon photonics platforms [9–12]. In this work, we demonstrate the first BTO-based 80 km coherent ZR modeled architecture and a 2 km bi-directional self-homodyne coherent-lite architecture, achieving net 1 Tbps, enabling high-speed inter and intra-data center networks.

2. Device Design and Manufacturing

The 3 mm BTO DP-IQM is fabricated by Lumiphase with a 200 mm BTO process [12]. An epitaxial BTO thin film is transferred from a silicon growth substrate onto a thermal-oxide wafer, after which high-speed and DC phase shifters are defined. The DC phase shifters replace on-chip heaters and leverage BTO's large Pockels coefficient for ultra-low-power DC tuning, where all DC phase shifters combined consume less than 5 mW under transmission conditions. Light is coupled from SMF to silicon nitride edge couplers via free-space optics incorporating polarization rotating and combining elements. 50 Ω terminations are integrated on-chip, and differential RF drivers with 23 dB gain are wire-bonded to the BTO dies true differential push-pull RF interface. The combined package, including the RF driver, BTO die, and test structures, exhibits a 1 GHz V_π of 0.4 V. The 6 dB electro-optic S21 of the DP-IQM package is ~ 25 GHz at both 30 and 85 $^\circ\text{C}$, and the fiber-to-fiber IL is 14.5 dB per polarization.

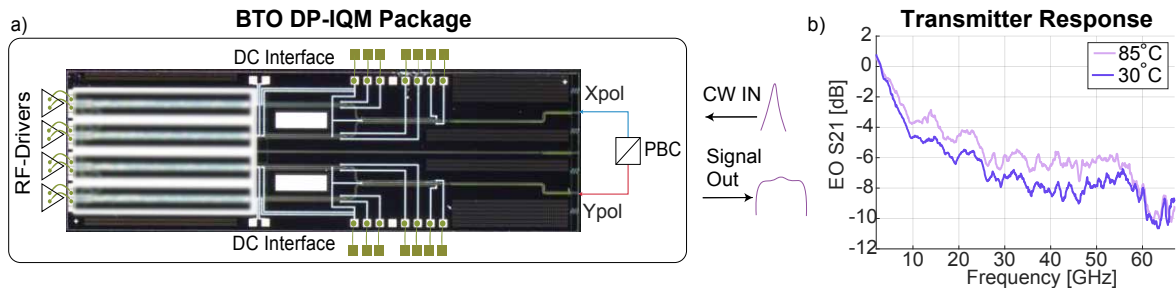


Fig. 1. a) Illustration of BTO DP-IQM package with a micrograph of a 3 mm BTO DP-IQM bare die, integrated differential RF drivers wire bonded to a GSSG RF interface with 23 dB gain, free-space optics for pol rotating and combining and separate wire bonds for DC interfacing. b) Electro-optic S21 of entire DP-IQM package including test structures, RF wire bonds, drivers and DP-IQM at 30 $^\circ\text{C}$ and 85 $^\circ\text{C}$.

3. Experimental Setup

Figure 2 a) shows the system modeled after the ZR architecture with an 80 km fiber span. Electrical symbols are generated by four differential channels of a 128 GSa/s M8199A arbitrary waveform generator (AWG). The data sent to each IQM is completely de-correlated. Offline Tx DSP is applied, where symbols are re-sampled to match the AWG sampling rate, root-raised cosine pulse shaped with a 5% roll-off, and pre-emphasized to compensate for the AWG, RF cables, and DP-IQM package. A C-band integrable tunable laser assembly (ITLA) with 15 dBm

output power and 100 kHz linewidth is used as both carrier and LO. The 80 km SMF span emulates a ZR link, where a booster and pre-amp EDFA compensate for fiber loss, package IL, and the absence of TIAs at the receiver. The signal is combined with an ASE loading path via a 3 dB coupler to sweep the OSNR as needed. The OSNR is measured using an OSA at the input to the 90° hybrid following the 12.5 GHz method. A DP 90° hybrid connects to four balanced photodiodes and two synchronized 63 GHz real-time oscilloscopes (RTOs) sampling at 160 GSa/s. Offline Rx DSP re-samples the waveform, applies CD and FO compensation, matched filtering, and a 151-tap 4×4 real-valued MIMO for equalization and polarization de-multiplexing.

In the coherent-lite system architecture, a single C-band ITLA with 15 dBm output power and 100 kHz linewidth serves as both carrier and LO through a 70/30 % power split. An EDFA compensates for the IL of the DP-IQM package and the absence of TIAs. Circulators are used to emulate a bi-directional duplex fiber system in which the LO counter-propagates the signal, where only one of the fibers in the duplex fiber is used [4]. This ensures that the signal and LO traverse the same distance. The fiber path lengths of the LO and signal are matched, which is important to mitigate laser phase noise in these types of systems [13]. Such duplex fiber configurations are commonly deployed in 2–5 km data center links, making the system practically feasible [2]. Because the LO traverses a fiber span, its polarization is rotated and must be re-aligned with that of the receiver. This is compensated using a three-paddle polarization controller, however, a low-speed analog compensator could be employed in practical systems [14]. The same transmitter DSP is applied as in the ZR system architecture. At the receiver, no CD compensation is applied for all SMF spans. Moreover, since one laser acts as both carrier and LO, FO compensation and carrier recovery DSP are omitted. A 151-tap 4×4 MIMO equalizer is employed for polarization de-multiplexing and to mitigate residual CD-induced ISI.

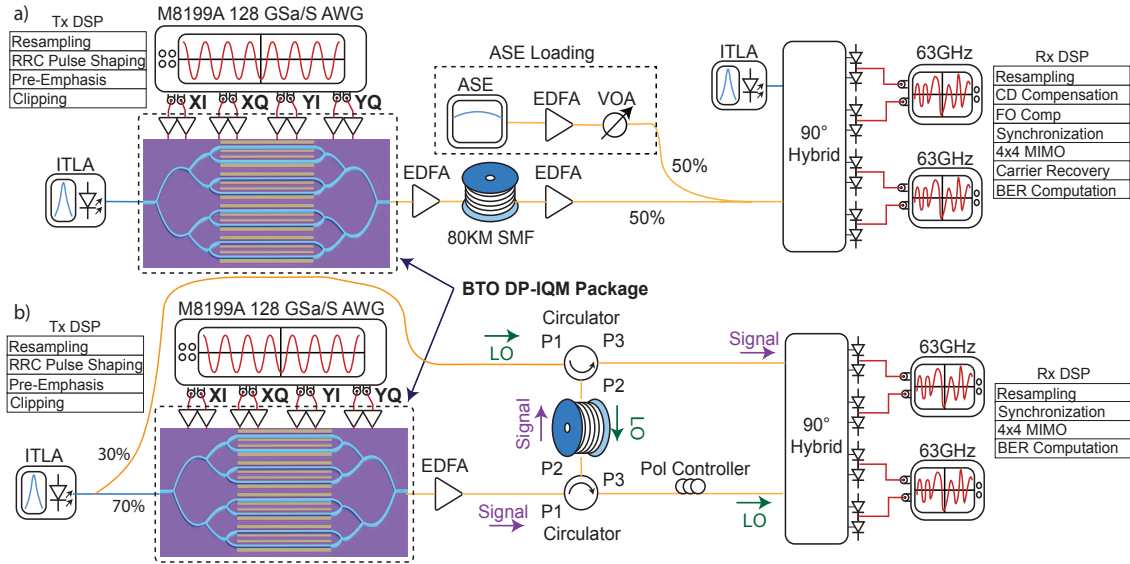


Fig. 2. Schematic of experimental setup and DSP stacks. Polarization is split and combined with free-space components in the package. a) ZR system architecture with an 80 km SMF span and 2 separate lasers for carrier and LO. b) Bi-directional self-homodyne coherent-lite system with fiber spans of 2/3/5 km SMF and a single laser for carrier and LO which no longer uses CD and FO comp or carrier recovery DSP.

4. Experimental Results

In the 80 km ZR system, results are shown in Fig. 3 a)–e). Net rates of 1 Tbps are achieved using 125 GBaud DP 32-QAM and 105 GBaud DP 64-QAM, both reaching bit error rates (BER) below the SD25 FEC threshold. Following the 800G ZR OIF standard, 120 GBaud DP 16-QAM achieves a BER of 6×10^{-3} , well below the OFEC threshold (2×10^{-2}). To verify high-temperature operation, the chip is heated to 85 °C (limited by the TEC), with no performance degradation observed, as shown in Fig. 3 b). Identical net rates and margins are maintained compared to room-temperature operation. BER is also measured versus received OSNR. For 120 GBaud DP 16-QAM, achieving a net 800 Gbps link under the OFEC threshold requires 28.74 dB OSNR, providing a 4 dB margin compared to the maximum achievable OSNR of 32.75 dB when driving the DP-IQM linearly. BER versus launch optical power (LOP) into the 80 km span shows an optimum LOP of 8 dBm, while the OFEC threshold is still met at -1.4 dBm. The square constellations exhibit similar error vector magnitude (EVM) across both polarizations.

In the bi-directional self-homodyne coherent-lite system, results are shown in Fig. 3 f)–h). Net rates of 1 Tbps are achieved using 125 GBaud DP 32-QAM and 105 GBaud DP 64-QAM, both below the SD25 FEC threshold. The 120 GBaud DP 16-QAM signal also meets the OFEC limit with a BER of 6.3×10^{-3} . The 2 km coherent-lite link exhibits performance comparable to the ZR system. Additional measurements at 3 km and 5 km using DP 16-QAM demonstrate that, without CD compensation DSP, the 5 km 120 GBaud DP 16-QAM signal remains recoverable under the OFEC threshold. System performance is also evaluated as a function of laser output power in a B2B configuration. A 2 dB reduction in output power renders the 120 GBaud DP 16-QAM signal unrecoverable

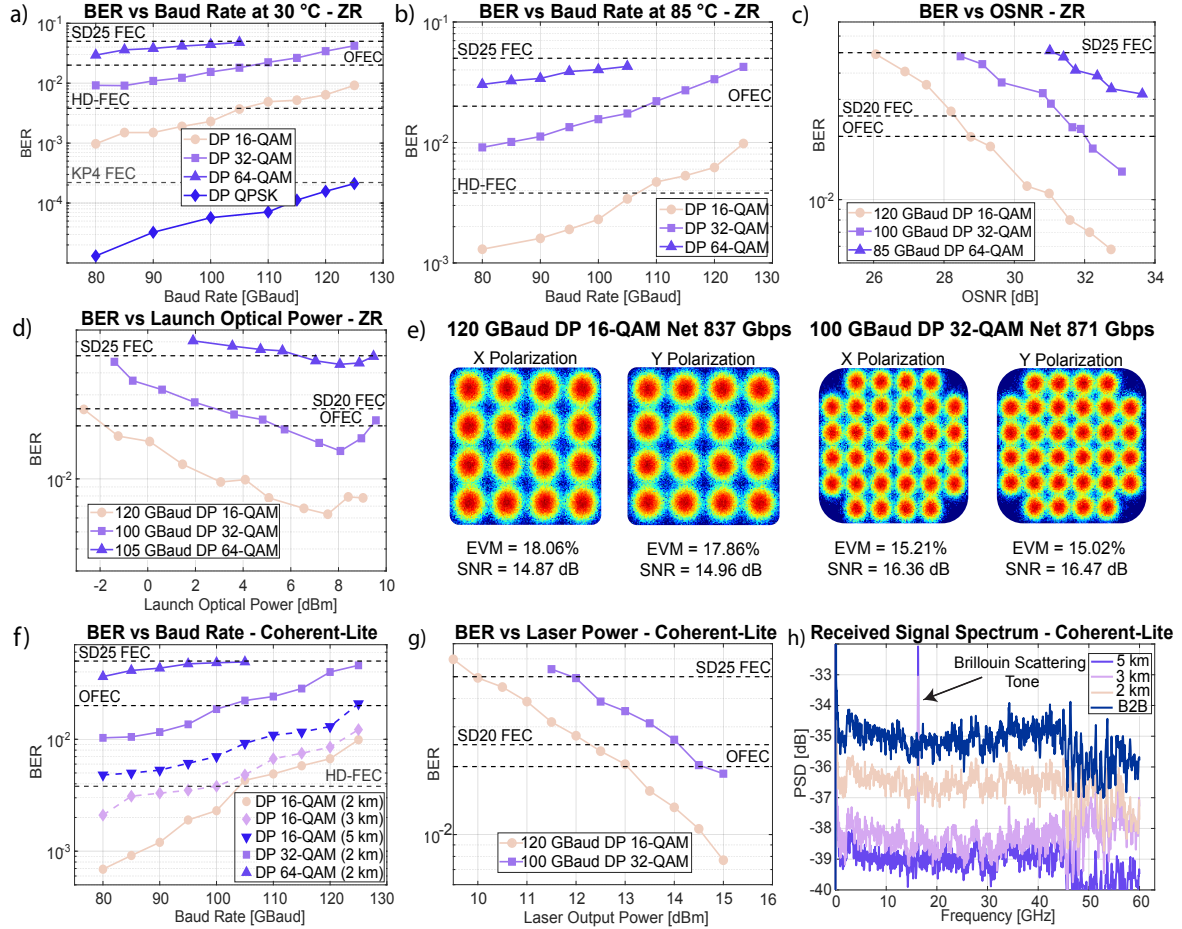


Fig. 3. ZR experimental results: a) BER vs baud rate at 30 °C. Net data rates of 1 Tbps are achieved at 125 Gbaud DP 32-QAM and 105 Gbaud DP 64-QAM b) BER vs baud rate at 85 °C where there is no system degradation compared to 30 °C. c) BER vs OSNR measured at the receiver. d) BER vs launch optical power into the 80km fiber. e) 120 Gbaud DP 16-QAM and 100 Gbaud DP 32-QAM constellations, EVM and SNR. Coherent-lite experimental results: f) BER vs baud rate. g) BER vs laser power measured B2B. Decrease in laser power affects both carrier and LO power. h) Signal spectrum measured at the RTO showing the Brillouin scattering generated frequency tone.

under the OFEC threshold, as the carrier and LO power decrease simultaneously. Although duplex fiber configurations make these types of systems practically feasible in data centers, the counter-propagating LO and signal introduce nonlinearities. As shown in Fig. 3 h), the received complex spectrum at the RTO exhibits a frequency tone at 17 GHz that strengthens with fiber length and is absent in B2B links. This tone arises from Brillouin scattering caused by a CW laser counter-propagating a signal through the fiber [15], representing a limiting factor for such bi-directional links. To prevent amplification of these unwanted frequency tones, the EDFA is placed at the transmitter. The degraded 5 km performance therefore results from the combined impact of reduced carrier and LO power, residual CD-induced ISI, and Brillouin scattering.

5. Conclusion

We demonstrate net 1 Tbps transmission over 80 km in a system modeled after a coherent ZR architecture, and over 2 km in a bi-directional self-homodyne coherent-lite system, reporting the highest data rate on BTO.

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